

# LJSF 3 REST API

## Complete v1 technical reference

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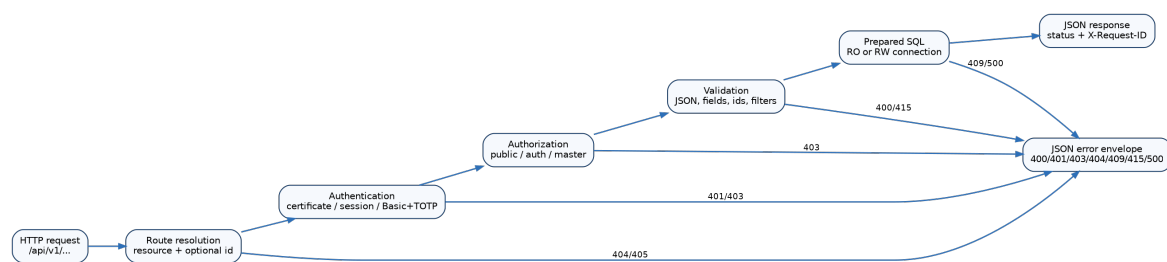


| Item                | Value                              |
|---------------------|------------------------------------|
| Product             | LJSF 3 - ATLAS Installation System |
| Application version | 3.0.0                              |
| Package revision    | r30                                |
| API version         | v1                                 |
| Base path           | /atlas_install/api/v1              |
| Format              | JSON / HTTPS                       |

Normative reference for REST API v1 included in package r30. Legacy endpoints remain available but are outside the REST contract.

# 1. Service identity

- **Product:** LJSF 3 - ATLAS Installation System
- **Application version:** 3.0.0
- **Package revision:** r30
- **Base path:** /atlas\_install/api/v1
- **Media type:** application/json; charset=UTF-8
- **Response cache:** Cache-Control: no-store
- **API version header:** X-API-Version: v1
- **Request correlation:** X-Request-ID

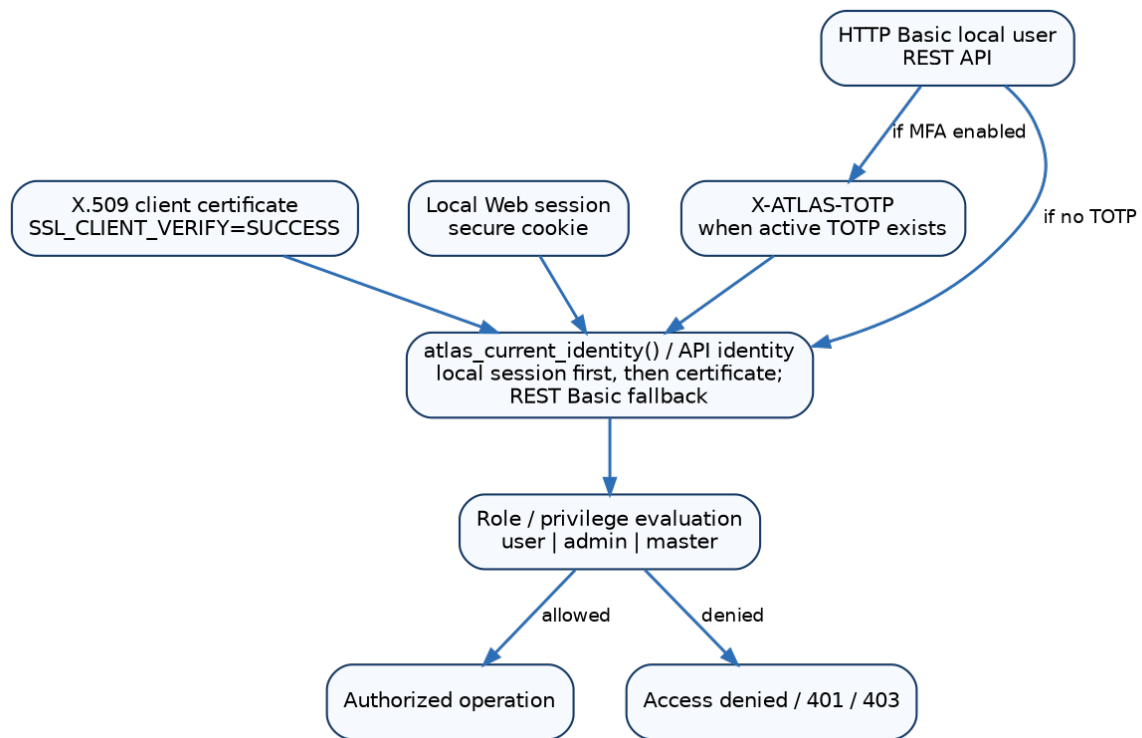


REST request lifecycle

Figure: REST request lifecycle.

# 2. Authentication model

The API reuses the application identity model. Identity resolution is: an existing local Web session; a valid X.509 client certificate; then, as a REST-specific fallback, HTTP Basic using a local account. If the local account has at least one active TOTP credential, Basic authentication also requires X-ATLAS-TOTP.



### Authentication model

| Mechanism         | Transport                    | Requirements                                                         | Notes                                             |
|-------------------|------------------------------|----------------------------------------------------------------------|---------------------------------------------------|
| X.509 certificate | TLS client certificate       | TLS reaches Apache;<br>SSL_CLIENT_VERIFY=SUCCESS; DN mapped in user. | Suitable for personal/robot certificates.         |
| Local session     | Secure Web cookie            | Unexpired local session and enabled account.                         | Useful for browser-originated API calls.          |
| HTTP Basic        | Authorization: Basic ...     | Enabled local account, valid password, must_change_password=0.       | Use only over HTTPS.                              |
| Basic + TOTP      | Basic + X-ATLAS-TOTP: 123456 | Required when at least one active TOTP exists.                       | Six digits; current TOTP slice +/- 1 is accepted. |

## 2.1 Authentication examples

```
# Basic without TOTP
curl --fail-with-body -u operator:'PASSWORD' \
  https://atlas-install.example.org/atlas_install/api/v1/me

# Basic with TOTP
curl --fail-with-body -u operator:'PASSWORD' \
  -H 'X-ATLAS-TOTP: 123456' \
  https://atlas-install.example.org/atlas_install/api/v1/me

# X.509
curl --fail-with-body --cert client.pem --key client.key \
  https://atlas-install.example.org/atlas_install/api/v1/me
```

## 2.2 Authentication failures

- `401 authentication_required`: no valid identity. Response includes `WWW-Authenticate: Basic realm="LJSF 3 REST API"`.
- `401 totp_required`: Basic credentials are valid but TOTP is missing/invalid.
- `403 password_change_required`: the local account must change its password through the Web UI before API use.
- `403 forbidden`: valid authentication but insufficient role.

## 3. HTTP and JSON conventions

| Item            | Behavior                                                                                              |
|-----------------|-------------------------------------------------------------------------------------------------------|
| GET collection  | 200 with <code>{data:[...], meta:{resource,total,limit,offset,count}}</code>                          |
| GET item        | 200 with <code>{data:{...}}</code> ; 404 if not found                                                 |
| POST collection | Requires JSON body; 201 Created; Location header; response contains created id and location           |
| PATCH item      | Partial update; 200 with the updated resource                                                         |
| PUT item        | Currently has the same supplied-field partial-update semantics as PATCH; it is not a full replacement |
| DELETE item     | 204 No Content; 404 if missing; 409 on database conflicts                                             |
| OPTIONS         | Returns supported method names; router advertises GET, HEAD, POST, PUT, PATCH, DELETE, OPTIONS        |
| HEAD            | Accepted on read routes; web server may suppress the body as required by HTTP                         |
| Content-Type    | POST/PUT/PATCH with a non-empty body require <code>application/json</code> ; otherwise 415            |

| Item                     | Behavior                                                      |
|--------------------------|---------------------------------------------------------------|
| Unknown JSON field       | 400 unknown_field; immutable keys are also rejected on update |
| Arrays/objects in fields | Rejected with 400 invalid_field; resource fields are scalar   |
| Caching                  | All API responses send Cache-Control: no-store                |

### 3.1 Error envelope

```
{
  "error": {
    "status": 400,
    "code": "invalid_request",
    "message": "...",
    "request_id": "7c3f2a9b...",
    "details": {}
  }
}
```

### 3.2 Status codes

| HTTP | Name                   | Meaning                                           |
|------|------------------------|---------------------------------------------------|
| 200  | OK                     | Successful read or update.                        |
| 201  | Created                | Successful creation; use Location.                |
| 204  | No Content             | Successful deletion.                              |
| 400  | Bad Request            | Invalid id/JSON/field/request.                    |
| 401  | Unauthorized           | Missing authentication or TOTP required.          |
| 403  | Forbidden              | Insufficient role or password change required.    |
| 404  | Not Found              | Unknown route/resource/record.                    |
| 405  | Method Not Allowed     | Method unsupported by route.                      |
| 409  | Conflict               | Database constraint/conflict during write/delete. |
| 415  | Unsupported Media Type | Non-JSON request body.                            |
| 500  | Internal Server Error  | Internal error; correlate request_id with logs.   |

## 4. Collections, filtering, search, pagination and sorting

Every exposed field can be supplied as a query parameter for exact matching ( `field=value` ). `q` performs `LIKE %q%` only on the resource search fields. `limit` defaults to 200 and is capped at 1000; `offset` cannot be negative. `sort` must be an exposed field or the primary key is used; `order` accepts `ASC` or `DESC` , otherwise `ASC` .

```
curl 'https://atlas-install.example.org/atlas_install/api/v1/releases?
      limit=100&offset=0&sort=name&order=DESC&q=24.3'
```

## 5. Resource matrix

| Resource              | DB table             | Key | GET                | Write              | fields                                             |
|-----------------------|----------------------|-----|--------------------|--------------------|----------------------------------------------------|
| releases              | release_data         | ref | public             | authenticated user | name, tag, package, comments                       |
| sites                 | site                 | ref | public             | authenticated user | cs, cename, name, atlas_name, alias, arch          |
| requests              | request              | id  | authenticated user | authenticated user | id, user_comments, admin_comments                  |
| tasks                 | task                 | ref | public             | authenticated user | name, description, target                          |
| architectures         | release_arch         | ref | public             | authenticated user | platform_type, os_type, gcc_ver, mode, description |
| targets               | autoinstall_target   | ref | public             | authenticated user | name, description                                  |
| infosys               | bdii                 | ref | public             | authenticated user | hostname, ns, lb, ld, wmpoxy, myproxy              |
| release-subscriptions | release_subscription | ref | authenticated user | authenticated user | sitename, pattern, comment                         |
| release-status        | release_stat         | ref | public             | authenticated user | name, tag, status, comments                        |

| Resource         | DB table         | Key | GET         | Write              | fields                                       |
|------------------|------------------|-----|-------------|--------------------|----------------------------------------------|
| grids            | grid             | ref | public      | authenticated user | name, description                            |
| facilities       | facility         | ref | public      | authenticated user | name, description                            |
| site-types       | site_type        | ref | public      | authenticated user | name, description                            |
| request-statuses | request_status   | ref | public      | master role        | description                                  |
| request-types    | request_type     | ref | public      | master role        | field, description, comment                  |
| users            | user             | ref | master role | master role        | name, dn, email                              |
| local-users      | atlas_local_user | id  | master role | master role        | username, first_name, last_name, email, role |

## 6. Special endpoints

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**GET** /

Returns API name, version, documentation URL and resource list.

**GET** /health

API health only; it is not the full application readiness probe. Response:

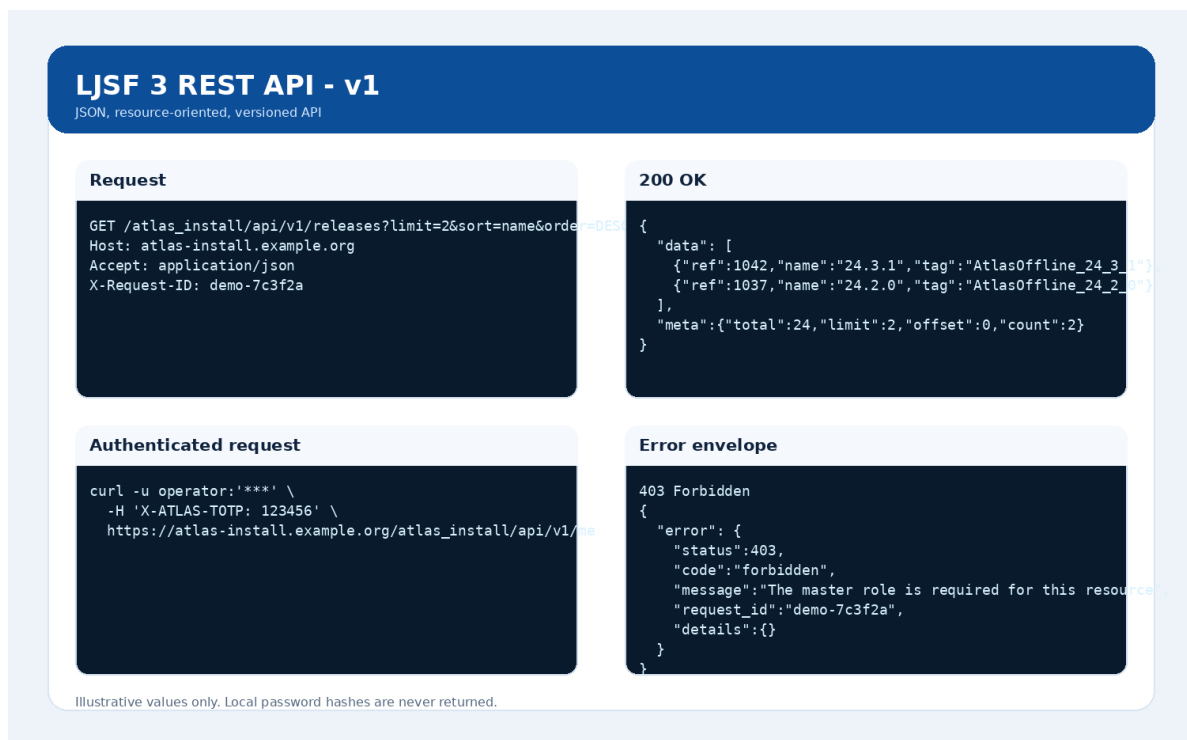
```
{"status": "ok", "api": "v1", "request_id": "..."}.
```

**GET** /me

Requires authentication and returns the effective identity. `password_hash` is always removed.

**GET** /openapi

Returns a compact OpenAPI 3.1 document with paths and methods; this manual is the more detailed source for fields, authentication and semantics.



API request/response screenshot

## 7. Complete resource reference

### 7.1 releases

**DB table:** `release_data`

**Key:** `ref` (integer)

**Read access:** public

**Write access:** authenticated user

**q search fields:** `name`, `tag`, `package`, `comments`

#### Methods:

- GET `/releases`
- GET `/releases/{id}`
- POST `/releases`
- PATCH `/releases/{id}`
- PUT `/releases/{id}`
- DELETE `/releases/{id}`

#### Exposed fields:

| Field             | DB type                  | Nullable | Writable | Description                  |
|-------------------|--------------------------|----------|----------|------------------------------|
| <code>ref</code>  | <code>int(11)</code>     | NO       | no       | Numeric resource identifier. |
| <code>name</code> | <code>varchar(50)</code> | NO       | yes      | Canonical or displayed name. |

| Field                 | DB type      | Nullable | Writable | Description                                                                      |
|-----------------------|--------------|----------|----------|----------------------------------------------------------------------------------|
| build                 | varchar(15)  | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| typefk                | int(11)      | NO       | yes      | Type reference.                                                                  |
| archfk                | int(11)      | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| sw_archfk             | int(11)      | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| userfk                | int(11)      | NO       | yes      | User reference.                                                                  |
| obsolete              | int(11)      | YES      | yes      | Obsolete-release flag.                                                           |
| autoinstall           | int(11)      | YES      | yes      | Automatic-installation flag.                                                     |
| requires              | varchar(30)  | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| dbrelease             | varchar(10)  | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| installer_version     | varchar(10)  | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| install_tools_version | varchar(10)  | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| sw_name               | varchar(255) | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |

| Field                  | DB type      | Nullable | Writable | Description                                                                      |
|------------------------|--------------|----------|----------|----------------------------------------------------------------------------------|
| sw_revision            | varchar(20)  | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| sw_physicalpath        | varchar(255) | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| sw_logicalpath         | varchar(255) | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| tag                    | varchar(100) | NO       | yes      | Software/release tag.                                                            |
| package                | varchar(255) | NO       | yes      | Software package or descriptor.                                                  |
| comments               | varchar(255) | YES      | yes      | Operational comments.                                                            |
| date                   | datetime     | NO       | yes      | Date/time associated with the record.                                            |
| cvmfs_available        | int(11)      | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| critical               | int(11)      | NO       | yes      | Critical-release flag.                                                           |
| critical_validity_from | datetime     | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| critical_validity_to   | datetime     | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| critical_description   | varchar(25)  | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/releases?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/releases/123
```

## 7.2 sites

**DB table:** site

**Key:** ref (integer)

**Read access:** public

**Write access:** authenticated user

**q search fields:** cs, cename, name, atlas\_name, alias, arch

### Methods:

- GET /sites
- GET /sites/{id}
- POST /sites
- PATCH /sites/{id}
- PUT /sites/{id}
- DELETE /sites/{id}

### Exposed fields:

| Field           | DB type      | Nullable | Writable | Description                                                                      |
|-----------------|--------------|----------|----------|----------------------------------------------------------------------------------|
| ref             | int(11)      | NO       | no       | Numeric resource identifier.                                                     |
| cs              | varchar(128) | NO       | yes      | Legacy computer-service identifier.                                              |
| cename          | varchar(50)  | NO       | yes      | Legacy CE name.                                                                  |
| name            | varchar(30)  | NO       | yes      | Canonical or displayed name.                                                     |
| atlas_name      | varchar(30)  | NO       | yes      | ATLAS site name.                                                                 |
| tier_level      | int(11)      | NO       | yes      | Site tier level.                                                                 |
| gridfk          | int(11)      | NO       | yes      | Grid reference.                                                                  |
| facilityfk      | int(11)      | YES      | yes      | Facility reference.                                                              |
| activity_typefk | int(11)      | YES      | yes      | Activity-type reference.                                                         |
| osname          | varchar(128) | NO       | yes      | OS name.                                                                         |
| osrelease       | varchar(128) | NO       | yes      | OS release.                                                                      |
| osversion       | varchar(128) | NO       | yes      | OS version.                                                                      |
| arch            | varchar(20)  | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |

| Field           | DB type      | Nullable | Writable | Description                                                                      |
|-----------------|--------------|----------|----------|----------------------------------------------------------------------------------|
| tags            | varchar(255) | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| alias           | varchar(128) | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| swarea          | varchar(255) | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| fstype          | varchar(15)  | YES      | yes      | Filesystem type.                                                                 |
| mountpoint      | varchar(255) | YES      | yes      | Mount point.                                                                     |
| capacity        | int(11)      | YES      | yes      | Capacity.                                                                        |
| available       | int(11)      | YES      | yes      | Available space.                                                                 |
| quota           | int(11)      | YES      | yes      | Quota.                                                                           |
| status          | int(11)      | YES      | yes      | Application status.                                                              |
| attr            | int(10)      | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| host_status     | int(11)      | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| resource_status | int(11)      | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| last_activity   | timestamp    | YES      | yes      | Last recorded activity.                                                          |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/sites?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/sites/123
```

## 7.3 requests

**DB table:** request

**Key:** id (string)

**Read access:** authenticated user

**Write access:** authenticated user

**q search fields:** id, user\_comments, admin\_comments

### Methods:

- GET /requests
- GET /requests/{id}
- POST /requests
- PATCH /requests/{id}
- PUT /requests/{id}
- DELETE /requests/{id}

### Exposed fields:

| Field          | DB type      | Nullable | Writable | Description                                                                      |
|----------------|--------------|----------|----------|----------------------------------------------------------------------------------|
| id             | varchar(128) | NO       | no       | Resource identifier.                                                             |
| bdiifk         | int(11)      | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| sitefk         | int(11)      | NO       | yes      | Site reference.                                                                  |
| relfk          | int(11)      | NO       | yes      | Release reference.                                                               |
| typefk         | int(11)      | NO       | yes      | Type reference.                                                                  |
| userfk         | int(11)      | NO       | yes      | User reference.                                                                  |
| adminfk        | int(11)      | YES      | yes      | Assigned administrator/operator reference.                                       |
| statusfk       | int(11)      | NO       | yes      | Request-status reference.                                                        |
| force_run      | int(11)      | NO       | yes      | Force-execution flag.                                                            |
| request_date   | datetime     | YES      | yes      | Request date/time.                                                               |
| update_date    | datetime     | YES      | yes      | Last update date/time.                                                           |
| user_comments  | varchar(255) | YES      | yes      | User comments.                                                                   |
| admin_comments | varchar(255) | YES      | yes      | Administrative comments.                                                         |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/requests?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/requests/REQ-EXAMPLE-001
```

## 7.4 tasks

**DB table:** task

**Key:** ref (integer)

**Read access:** public

**Write access:** authenticated user

**q search fields:** name, description, target

### Methods:

- GET /tasks
- GET /tasks/{id}
- POST /tasks
- PATCH /tasks/{id}
- PUT /tasks/{id}
- DELETE /tasks/{id}

### Exposed fields:

| Field       | DB type      | Nullable | Writable | Description                  |
|-------------|--------------|----------|----------|------------------------------|
| ref         | int(11)      | NO       | no       | Numeric resource identifier. |
| name        | varchar(50)  | NO       | yes      | Canonical or displayed name. |
| description | varchar(255) | YES      | yes      | Human-readable description.  |
| target      | varchar(20)  | YES      | yes      | Task target.                 |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/tasks?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/tasks/123
```

## 7.5 architectures

**DB table:** release\_arch

**Key:** ref (integer)

**Read access:** public

**Write access:** authenticated user

**q search fields:** platform\_type, os\_type, gcc\_ver, mode, description

### Methods:

- GET /architectures
- GET /architectures/{id}

- `POST /architectures`
- `PATCH /architectures/{id}`
- `PUT /architectures/{id}`
- `DELETE /architectures/{id}`

#### Exposed fields:

| Field         | DB type      | Nullable | Writable | Description                  |
|---------------|--------------|----------|----------|------------------------------|
| ref           | int(11)      | NO       | no       | Numeric resource identifier. |
| platform_type | varchar(16)  | NO       | yes      | Platform type.               |
| os_type       | varchar(128) | NO       | yes      | Operating-system type.       |
| gcc_ver       | varchar(128) | NO       | yes      | GCC/toolchain version.       |
| mode          | varchar(10)  | NO       | yes      | Build mode.                  |
| description   | varchar(255) | NO       | yes      | Human-readable description.  |

#### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/architectures?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/architectures/123
```

## 7.6 targets

**DB table:** `autoinstall_target`

**Key:** `ref` (integer)

**Read access:** public

**Write access:** authenticated user

**q search fields:** `name`, `description`

#### Methods:

- `GET /targets`
- `GET /targets/{id}`
- `POST /targets`
- `PATCH /targets/{id}`
- `PUT /targets/{id}`
- `DELETE /targets/{id}`

#### Exposed fields:

| Field | DB type | Nullable | Writable | Description                  |
|-------|---------|----------|----------|------------------------------|
| ref   | int(11) | NO       | no       | Numeric resource identifier. |

| Field       | DB type      | Nullable | Writable | Description                  |
|-------------|--------------|----------|----------|------------------------------|
| name        | varchar(15)  | NO       | yes      | Canonical or displayed name. |
| description | varchar(255) | YES      | yes      | Human-readable description.  |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/targets?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/targets/123
```

## 7.7 infosys

**DB table:** bdii

**Key:** ref (integer)

**Read access:** public

**Write access:** authenticated user

**q search fields:** hostname, ns, lb, ld, wmpoxy, myproxy

### Methods:

- GET /infosys
- GET /infosys/{id}
- POST /infosys
- PATCH /infosys/{id}
- PUT /infosys/{id}
- DELETE /infosys/{id}

### Exposed fields:

| Field      | DB type      | Nullable | Writable | Description                                                                      |
|------------|--------------|----------|----------|----------------------------------------------------------------------------------|
| ref        | int(11)      | NO       | no       | Numeric resource identifier.                                                     |
| facilityfk | int(11)      | NO       | yes      | Facility reference.                                                              |
| hostname   | varchar(128) | NO       | yes      | Host name.                                                                       |
| port       | int(11)      | NO       | yes      | TCP port.                                                                        |
| ns         | varchar(255) | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| lb         | varchar(255) | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |

| Field     | DB type      | Nullable | Writable | Description                                                                      |
|-----------|--------------|----------|----------|----------------------------------------------------------------------------------|
| Id        | varchar(255) | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| wmproxy   | varchar(255) | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| myproxy   | varchar(255) | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| preferred | int(11)      | NO       | yes      | Preference flag.                                                                 |
| enabled   | int(11)      | NO       | yes      | Enable flag (0/1).                                                               |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/infosys?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/infosys/123
```

## 7.8 release-subscriptions

**DB table:** release\_subscription

**Key:** ref (integer)

**Read access:** authenticated user

**Write access:** authenticated user

**q search fields:** sitename, pattern, comment

### Methods:

- GET /release-subscriptions
- GET /release-subscriptions/{id}
- POST /release-subscriptions
- PATCH /release-subscriptions/{id}
- PUT /release-subscriptions/{id}
- DELETE /release-subscriptions/{id}

### Exposed fields:

| Field    | DB type     | Nullable | Writable | Description                  |
|----------|-------------|----------|----------|------------------------------|
| ref      | int(11)     | NO       | no       | Numeric resource identifier. |
| sitename | varchar(50) | NO       | yes      | Site name.                   |

| Field   | DB type      | Nullable | Writable | Description                                                                      |
|---------|--------------|----------|----------|----------------------------------------------------------------------------------|
| userfk  | int(11)      | NO       | yes      | User reference.                                                                  |
| pattern | varchar(50)  | NO       | yes      | Subscription pattern.                                                            |
| date    | datetime     | NO       | yes      | Date/time associated with the record.                                            |
| comment | varchar(255) | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/release-subscriptions?
    limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/release-subscriptions/123
```

## 7.9 release-status

**DB table:** release\_stat

**Key:** ref (integer)

**Read access:** public

**Write access:** authenticated user

**q search fields:** name, tag, status, comments

### Methods:

- GET /release-status
- GET /release-status/{id}
- POST /release-status
- PATCH /release-status/{id}
- PUT /release-status/{id}
- DELETE /release-status/{id}

### Exposed fields:

| Field     | DB type     | Nullable | Writable | Description                  |
|-----------|-------------|----------|----------|------------------------------|
| ref       | int(11)     | NO       | no       | Numeric resource identifier. |
| sitefk    | int(11)     | NO       | yes      | Site reference.              |
| name      | varchar(50) | NO       | yes      | Canonical or displayed name. |
| userfk    | int(11)     | NO       | yes      | User reference.              |
| pin       | int(11)     | YES      | yes      | Pin flag.                    |
| pinuserfk | int(11)     | YES      | yes      | User who set the pin.        |

| Field    | DB type      | Nullable | Writable | Description                           |
|----------|--------------|----------|----------|---------------------------------------|
| pindate  | datetime     | YES      | yes      | Pin date.                             |
| tag      | varchar(100) | NO       | yes      | Software/release tag.                 |
| status   | varchar(100) | NO       | yes      | Application status.                   |
| comments | varchar(255) | YES      | yes      | Operational comments.                 |
| date     | datetime     | NO       | yes      | Date/time associated with the record. |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/release-status?
    limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/release-status/123
```

## 7.10 grids

**DB table:** grid

**Key:** ref (integer)

**Read access:** public

**Write access:** authenticated user

**q search fields:** name, description

### Methods:

- GET /grids
- GET /grids/{id}
- POST /grids
- PATCH /grids/{id}
- PUT /grids/{id}
- DELETE /grids/{id}

### Exposed fields:

| Field       | DB type      | Nullable | Writable | Description                  |
|-------------|--------------|----------|----------|------------------------------|
| ref         | int(11)      | NO       | no       | Numeric resource identifier. |
| name        | varchar(10)  | NO       | yes      | Canonical or displayed name. |
| description | varchar(255) | YES      | yes      | Human-readable description.  |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/grids?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/grids/123
```

## 7.11 facilities

**DB table:** facility

**Key:** ref (integer)

**Read access:** public

**Write access:** authenticated user

**q search fields:** name, description

### Methods:

- GET /facilities
- GET /facilities/{id}
- POST /facilities
- PATCH /facilities/{id}
- PUT /facilities/{id}
- DELETE /facilities/{id}

### Exposed fields:

| Field       | DB type      | Nullable | Writable | Description                  |
|-------------|--------------|----------|----------|------------------------------|
| ref         | int(11)      | NO       | no       | Numeric resource identifier. |
| name        | varchar(10)  | NO       | yes      | Canonical or displayed name. |
| description | varchar(255) | YES      | yes      | Human-readable description.  |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/facilities?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/facilities/123
```

## 7.12 site-types

**DB table:** site\_type

**Key:** ref (integer)

**Read access:** public

**Write access:** authenticated user

**q search fields:** name, description

### Methods:

- GET /site-types
- GET /site-types/{id}
- POST /site-types

- PATCH /site-types/{id}
- PUT /site-types/{id}
- DELETE /site-types/{id}

#### Exposed fields:

| Field       | DB type      | Nullable | Writable | Description                  |
|-------------|--------------|----------|----------|------------------------------|
| ref         | int(11)      | NO       | no       | Numeric resource identifier. |
| name        | varchar(50)  | NO       | yes      | Canonical or displayed name. |
| description | varchar(255) | NO       | yes      | Human-readable description.  |

#### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/site-types?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/site-types/123
```

## 7.13 request-statuses

**DB table:** request\_status

**Key:** ref (integer)

**Read access:** public

**Write access:** master role

**q search fields:** description

#### Methods:

- GET /request-statuses
- GET /request-statuses/{id}
- POST /request-statuses
- PATCH /request-statuses/{id}
- PUT /request-statuses/{id}
- DELETE /request-statuses/{id}

#### Exposed fields:

| Field       | DB type      | Nullable | Writable | Description                  |
|-------------|--------------|----------|----------|------------------------------|
| ref         | int(11)      | NO       | no       | Numeric resource identifier. |
| description | varchar(128) | YES      | yes      | Human-readable description.  |

#### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/request-statuses?
    limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/request-statuses/123
```

## 7.14 request-types

**DB table:** request\_type

**Key:** ref (integer)

**Read access:** public

**Write access:** master role

**q search fields:** field, description, comment

### Methods:

- GET /request-types
- GET /request-types/{id}
- POST /request-types
- PATCH /request-types/{id}
- PUT /request-types/{id}
- DELETE /request-types/{id}

### Exposed fields:

| Field       | DB type      | Nullable | Writable | Description                                                                      |
|-------------|--------------|----------|----------|----------------------------------------------------------------------------------|
| ref         | int(11)      | NO       | no       | Numeric resource identifier.                                                     |
| level       | int(11)      | NO       | yes      | Level/precedence.                                                                |
| field       | varchar(100) | NO       | yes      | Application field name.                                                          |
| description | varchar(128) | YES      | yes      | Human-readable description.                                                      |
| comment     | varchar(255) | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/request-types?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/request-types/123
```

## 7.15 users

**DB table:** user

**Key:** ref (integer)

**Read access:** master role

**Write access:** master role

**q search fields:** name, dn, email

**Methods:**

- GET /users
- GET /users/{id}
- POST /users
- PATCH /users/{id}
- PUT /users/{id}
- DELETE /users/{id}

**Exposed fields:**

| Field       | DB type      | Nullable | Writable | Description                                                                      |
|-------------|--------------|----------|----------|----------------------------------------------------------------------------------|
| ref         | int(11)      | NO       | no       | Numeric resource identifier.                                                     |
| name        | varchar(50)  | NO       | yes      | Canonical or displayed name.                                                     |
| dn          | varchar(200) | YES      | yes      | X.509 Distinguished Name.                                                        |
| email       | varchar(150) | YES      | yes      | Associated email address.                                                        |
| priv_view   | int(11)      | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| priv_insert | int(11)      | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| priv_update | int(11)      | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| priv_pin    | int(11)      | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| priv_relsb  | int(11)      | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |

| Field         | DB type  | Nullable | Writable | Description                                                                      |
|---------------|----------|----------|----------|----------------------------------------------------------------------------------|
| priv_critical | int(11)  | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| rolefk        | int(11)  | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| valid_start   | datetime | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| valid_end     | datetime | YES      | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| enabled       | int(11)  | NO       | yes      | Enable flag (0/1).                                                               |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/users?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/users/123
```

## 7.16 local-users

**DB table:** atlas\_local\_user

**Key:** id (integer)

**Read access:** master role

**Write access:** master role

**q search fields:** username, first\_name, last\_name, email, role

### Methods:

- GET /local-users
- GET /local-users/{id}
- POST /local-users
- PATCH /local-users/{id}
- PUT /local-users/{id}
- DELETE /local-users/{id}

### Exposed fields:

| Field | DB type | Nullable | Writable | Description          |
|-------|---------|----------|----------|----------------------|
| id    | BIGINT  | NO       | no       | Resource identifier. |

| Field                | DB type      | Nullable | Writable | Description                                                                      |
|----------------------|--------------|----------|----------|----------------------------------------------------------------------------------|
| username             | VARCHAR(64)  | NO       | yes      | Local account name.                                                              |
| first_name           | VARCHAR(100) | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| last_name            | VARCHAR(100) | NO       | yes      | Canonical application-schema field; operational meaning depends on the resource. |
| email                | VARCHAR(254) | NO       | yes      | Associated email address.                                                        |
| role                 | VARCHAR(32)  | NO       | yes      | Local application role: user, admin, or master.                                  |
| enabled              | TINYINT(1)   | NO       | yes      | Enable flag (0/1).                                                               |
| must_change_password | TINYINT(1)   | NO       | yes      | When 1, password change is required before API use.                              |
| created_at           | DATETIME     | NO       | yes      | Creation date/time.                                                              |
| updated_at           | DATETIME     | NO       | yes      | Last update date/time.                                                           |

### Examples:

```
curl https://atlas-install.example.org/atlas_install/api/v1/local-users?limit=20&offset=0
curl https://atlas-install.example.org/atlas_install/api/v1/local-users/123
```

`password` is a special input-only field: accepted by POST/PATCH/PUT, minimum 8 characters, converted into `password_hash`, and never returned.

## 8. End-to-end examples

---

### 8.1 Create and update a local user

```
curl -u admin:'ADMIN_PASSWORD' -H 'Content-Type: application/json' \
-d '{"username":"operator","first_name":"API","last_name":"Operator","email":"operator@example.org","role":"user","enabled":1,"password":"StrongPass-2026"}' \
https://atlas-install.example.org/atlas_install/api/v1/local-users

curl -u admin:'ADMIN_PASSWORD' -X PATCH -H 'Content-Type: application/json' \
-d '{"role":"admin","must_change_password":0}' \
https://atlas-install.example.org/atlas_install/api/v1/local-users/5
```

### 8.2 Python

```
import requests

base = 'https://atlas-install.example.org/atlas_install/api/v1'
r = requests.get(f'{base}/releases', params={'q':'24.3','limit':50}, timeout=30)
r.raise_for_status()
for release in r.json()['data']:
    print(release['ref'], release['name'], release['tag'])
```

### 8.3 Robust error handling

```
import requests

r = requests.get('https://atlas-install.example.org/atlas_install/api/v1/requests',
                 timeout=30)
if r.status_code >= 400:
    payload = r.json()
    err = payload.get('error', {})
    print('HTTP', r.status_code, err.get('code'), err.get('message'), 'request_id=',
          err.get('request_id'))
```

## 9. Security, compatibility and current limitations

---

- Always use HTTPS; Basic sends reusable credentials on every request.
- Local passwords/hashes are never returned.
- Writes use parameterized queries and an exposed-field allowlist.
- There are currently no ETag/If-Match semantics and no Idempotency-Key header: clients must avoid blind POST retries.
- There are no bulk endpoints: each request modifies one resource.
- PUT is intentionally patch-like in v1; do not assume full replacement semantics.
- Maximum page size is 1000; iterate with offset for larger datasets.

- Resources with DB dependencies may return 409 on DELETE/UPDATE due to constraints/conflicts.
- Legacy endpoints are separate; do not infer REST contracts from legacy endpoint syntax.

## 10. Production client checklist

---

- ☐ Verify the server TLS certificate.
- ☐ Set explicit client-side timeouts.
- ☐ Record and propagate `X-Request-ID` where possible.
- ☐ Handle 401, 403, 409, 415 and 5xx separately.
- ☐ Never log passwords, Basic Authorization or TOTP codes.
- ☐ Use pagination and reasonable limits.
- ☐ Before DELETE, account for dependencies and handle 409.
- ☐ Use dedicated automation identities with the least required role.